Argument Principle.

Let f be a non-constant function analytic at z_0 with $f(z_0)=0$.
Then, for some $r>0$, f have only zero as z_0 in $B_r(z_0)$. Thus,

$$f(z)=(z-z_0)^k \cdot F(z) \text{ for some } k\in\mathbb{N},$$

and analytic function F at z_0 with $F(z_0)\neq 0$. Now,

$$f'(z)=k(z-z_0)^{k-1}F(z)+(z-z_0)^k\cdot F'(z),$$

this implies $\dfrac{f'(z)}{f(z)}=\dfrac{k}{z-z_0}+\dfrac{F'(z)}{F(z)}$.

Since $F(z_0)\neq 0$, f'/f has simple pole at z_0, with Residue k. So,

$$\int_C \frac{f'(z)}{f(z)}dz = 2\pi i\cdot k.$$

Now, let f has a pole of order n at z_0, then

$$f(z)=\frac{F(z)}{(z-z_0)^n}$$

where F is analytic at z_0 and $F(z_0)\neq 0$. Then,

$$f'(z)=-n\cdot\frac{F(z)}{(z-z_0)^{n+1}}+\frac{F'(z)}{(z-z_0)^n},$$

So therefore $\dfrac{f'(z)}{f(z)}=\dfrac{(z-z_0)^n}{F(z)}\cdot\left(-n\cdot\dfrac{F(z)}{(z-z_0)^{n+1}}+\dfrac{F'(z)}{(z-z_0)^n}\right)=\dfrac{-n}{z-z_0}+\dfrac{F'(z)}{F(z)}$

So,

$$\int_C \frac{f'(z)}{f(z)} = 2\pi i\cdot(-n).$$

obtain the two results, we obtain the following:

Thm. Let f be analytic inside and on a simple closed contour C except for finite poles in C, and $f(z)\neq 0$ on C. Then, counted with multiplicity, numbers of zeros N_f and poles P_f,

$$\frac{1}{2\pi i}\cdot\int_C \frac{f'(z)}{f(z)}dz = N_f - P_f.$$

Example.

Let $f(z)=\dfrac{(z+1)(z+7)^5(z-i)^2}{(z^2-2z+2)^4(z+i)^8(z-5i)^8}$ and $C: |z|=2$.

Then, $z=-1, i, i$ are zeros of f inside C,

and $z^2-2z+2=(z-1)^2+1$, so $1\pm i$ and $-i$ are poles of order of 8 and 8.

Therefore, $\displaystyle\int_{|z|=2}\frac{f'(z)}{f(z)}\,dz=2\pi i(3-16)=-26\pi i.$

Thm. Rouché Theorem.

Let f and g be analytic inside and on a simple closed countor C, with

$$|g(z)| < |f(z)| \text{ for all } z \in C.$$

Then, $f+g$ and f have the same number of zeros inside C.

Example.

Let $P(z) = z^{10} - 6z^9 - 3z + 1$. Put $f(z) = -6z^9 + 1$ and $g(z) = z^{10} - 3z$.

Then, $|f(z)| = |-6z^9 + 1| \geq 6 - 1 = 5 \geq 1 + 3 \geq |z^{10} - 3z| = |g(z)|$ on $|z| = 1$.

So, $P = f+g$ and f has number of roots in $|z|=1$, nine roots.

Example.

Let $P(z) = z^5 + 6z^3 + 2z + 10$. Put $f(z) = z^5 + 6z^3 + 2z$, $g(z) = 10$. Then, on $|z| = 1$,

$|f(z)| = 9 < 10 = |g(z)|$, so $f+g$ has no root in $|z|=1$ same as g

And, on $|z|=1$, $|P(z)| \geq 10 - 1 - 6 - 2 \geq 1$, so $P(z) \neq 0$ on $|z| = 1$.

And, let $f(z) = z^5$ and $g(z) = 6z^3 + 2z + 10$. Then, on $|z| = 3$,

$$3^5 \geq 6 \cdot 3^3 + 2 \cdot 3 + 10,$$

So all five zeros of P lie in $1 < |z| < 3$.